Modeling and Prediction of Bus Operation States for Bunching Analysis

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ABSTRACT

Bus bunching deteriorates transit service quality and passenger's experience. The modeling and prediction of bus operation states are essential for improving transit quality of service. Due to the nature of traffic evolution and state transition, bunching-oriented modeling based on bus operation state is more intuitive when compared with headway-based modeling approach. This manuscript explicitly predicted bus operation state by modeling the dynamic evolutions between different states. Five different bus operation states were defined and classified by K-means algorithm, and the dynamic state evolution was formulated as a Markov chain model. Finally, a multinomial logistic model was developed to predict the bus operation state. A case study was designed to test the performance of the proposed model, based on the GPS trajectory data collected from four bus routes in Xi'an, China. The results showed that the proposed model was able to accurately predict the bus operation states.

Keywords: public transit; bus operation state prediction; bus bunching; headway variation; Markov chain;

Introduction

Under ideal conditions, transit vehicles arrive at bus stations at a predetermined time following the established timetable. The term “bus bunching” refers to the phenomenon that buses, in a single-line, start by following a predetermined timetable, however as time goes by, two or more buses will run together or arrive at a station at the same time. Such abnormal changes of bus operation state can significantly affect the efficiency, quality of service (Wei and Migule 2011; Zhang et al. 2011; Sirmatel et al. 2017) and passenger satisfaction (de Oña et al. 2014; Chico et al. 2018) and lead to increased operating costs of the public transit system (Hammerle et al. 2005; Furth et al. 2007). As such, it is essential to model and predict real-time bus operation state to evaluate transit service quality for better transit system management and control purposes.Transit quality of service is affected by many factors. For example, the bus operation time was considered to be a key factor affecting bus quality (Li et al. 2013), and thus, it had often been used to evaluate the bus route operation state (Strathman et al. 2002; Mazloumi et al. 2008; El - Geneidy et al. 2011; Yan et al. 2016). Meanwhile, the operation speed was also widely adopted by researchers (Hans et al. 2015; Yan et al. 2016; Cortés et al. 2011; Sirmatel et al. 2018). With the outbreak of big data, temporal and spatial distributions based on time and speed had also been explored to further characterize the bus operation state (Yan et al. 2016; Cortés et al. 2011; Sirmatel et al. 2018; Yuan et al. 2015). From the perspective of passengers, variables such as load rate and average waiting time were used to identify the congestion in the bus line for control (Wang et al. 2017). Further, the interactions between buses, passengers and cars on a bus route were modelled to evaluate bus operation state (Dimitrov et al. 2017). Other additional indicators had also been proposed to explore the bus operation state, for example, Yan et al. (2016) selected percentile travel times, coefficient of variation of travel times, average speed and travel time distribution. Wang et al. (2017) considered transfer distance and bus punctual rate, and two level fuzzy analytic hierarchy processes were proposed to classify bus operation state as six levels. Deng et al. (2020) studied the impact of heterogenous roadway condition and intersection delay on bus bunching occurance.

The study of Xiao and Cheng (2015) divided the bus state into five categories, including normal, empty, bus failure, congestion and bunching.

The bus headway deviation contributes to the reliability of a bus route (Ding et al. 2009; Ruan 2009; Ruan and Lin 2009). Therefore, many existing researches employed headway, which was defined as the operation time difference between two consecutive buses, to explore the bus operation state. For example, Song et al. (2013) and Zhang et al. (2015) defined bus operation states based on headway deviation. Variations of headways were adopted to evaluate high frequency bus operation state, which were classified into six service levels (TRB. 2013). Existing research typically focused on the prediction of bus headways. For example, the headway fluctuation at the subsequent stops was modeled, and a least square support vector machine regression was established for headway prediction in (Wang 2014). Andres proposed an analytical model of an artificial bus route to predict the headway (Andres and Nair 2017), in which a shrinking horizon model predictive controller (MPC) was proposed to predict the short time-space behavior of buses so that the arrival time could be obtained. Moreira-Matias et al. (2014) used both historical and real-time data to predict the headway distributions at further stops.

In this manuscript, we argue that the bus operation state (e.g. bus bunching state) is a more intuitive and easy-to-understand performance measure than headway deviations. Bus operation state is important not only to the agencies and passengers, but also because the operation state evaluation based on the degree of bus bunching is more intuitive than headway, especially when the departure interval of the buses varies. Further, due to the dynamic nature of the traffic, different bus operation states can be observed and defined and found to transition from one state to another, i.e. the state evolution can be observed with certain transition probabilities. As such, when compared with the existing research that was usually based on headway deviation, we argue that the prediction of bus operation state has research significance and deserves more research attentions.

This manuscript explicitly predicted the bus operation state by modeling its dynamic evolutions between different states. Five different bus operation states were defined and classified based on K-means algorithm. The operation states were found to evolve from one to another with certain transition

probabilities that can be defined and calculated. The dynamic state evolution was formulated as a Markov chain model. Finally, a multinomial logistic model was developed to predict the bus operation state. A case study was designed to test the performance of the proposed model based on the GPS trajectory data collected from four bus routes in Xi'an, China. The results showed that the proposed model was able to accurately predict the bus operation states.

The rest of the paper is organized as follows: Section 2 presents the methodology developed to model the dynamic evolution of bus operation state and subsequently, the prediction model. A case study is presented in section 3 to implement the proposed model and examine its performance. Finally, Section 4 concludes manuscript findings and discusses future work.

Methodology

For convenient reference, the mathematical notations used in this section are presented below.

i: bus trip index, $i \in [1, 2, 3, 4 \dots \dots]$

j: bus stop index, $j \in [1, 2, 3, 4 \dots \dots]$

m: number for the studied bus routes. $m = \{\#600, \#601, \#308, \#206\}$

c: bus operation state

C: set of bus operation states, including the following states, 1 bus bunching, 2: bunching transition, 3: normal operation, 4: large interval transition, and 5: large interval

$c_{m,j}^{i}$: bus operation state for the ith trip of bus route m from stop $j-1$ to stop j

$H_{m,c}^{-}$: lower limit for the state c of bus route m

$H_{m,c}^{+}$: upper limit for the state c of bus route m

$h_{m,j}^{i}$: bus headway for the route m at stop j of ith bus trip

F_{kl}^{m}: frequency of state transition from state k to l after one step transition of route m

F_{k}^{m}: frequency of state k occurrence of route m

p_{kl}^{m}: one step transition probability from state k to state l of route m

$P_{(1)}^{m}$: one-step transition probability matrix of route m

$P_{(n)}^{m}$: n-step transition probability matrix of route m

State definition and clustering

In this study, headway was adopted to evaluate the microscopic bus operation state, due to the reason that headway can not only measure the distance between two consecutive buses but was also able to directly identify bus bunching phenomenon. Headway deviation was another related and commonly used variable for bus operation state identification. Headway deviation measured the fluctuation of the headway, and a higher deviation value indicated the bus is operating significant differently from its original schedule, and the chances of bus bunching happening will also become higher. However, headway deviation as an absolute value, may not be a good variable to capture bus bunching. For example, two buses may exhibit same headway deviation, but their headways could be completely different, and thus may have different implications to the bus operation state.

Consider a simple scenario as shown in Figure 1, where only two transit vehicles were considered as shown. At the beginning the headway between the two buses equaled to the scheduled headway, and thus the state was normal. As illustrated in case 1 on the left side, if due to whatever reason the leading bus was delayed, it would gather more passengers at the subsequent stops as the passengers were being accumulated. However, the second bus would encounter fewer customers due to the shorter headway between them. In other words, the initial disturbance would make the leading bus run slower while the second bus faster, until these two buses eventually bunched together. Another scenario was illustrated on the right side. If the second bus was delayed, it will run even slower as the subsequent bus stations will accumulate more passengers due to the larger headway, whereas the leading bus was not affected and will continue to operate according to the original schedule. As time goes, a large headway deviation will be observed eventually.

[Insert Figure 1: Bus operation state illustration Here]

Following such analysis, the microscopic bus operation state can be classified into five different categories: bus bunching, bunching transition, normal operation, large interval transition and large interval.

Therefore, the bus running state set can be expressed as $C = \{1,2,3,4,5\}$ with 1 for bus bunching, 2 for bunching transition, 3 for normal operation, 4 for large interval transition, and 5 for large interval.

As a commonly used method to classify the untrained sample into several groups, K-means clustering algorithm (Borgwardt et al. 2017) was employed for the classification of microscopic bus operation state based on the headway indicator. The goal was to divide the headway data into several disjoint clusters, with each representing a different underlying state. The Euclidean distance was used to calculate the distance between two different objects. K was set to be 5 and thus five cluster centers were obtained after the iteration process ended. When the five cluster centers were arranged in increasing order, the average of the adjacent cluster center was set to be the threshold of each state.

Modeling of dynamic state evolution

Probability function of headway distribution

In order to quantitatively identify the bus running states, the probability of bus line m running in a certain state at site j is defined as follows:

$$P_{m,j}^{c} = P(H_{m,c}^{-} \leq h_{m,j} \leq H_{m,c}^{+}) \tag{1}$$

Where $c \in C$ is the bus operation state, $H_{m,c}^{-}$ and $H_{m,c}^{+}$ are the lower and upper limits of state c, and $h_{m,j}$ is the bus headway values of route m at stop j.

If the headway probability distribution function of route m is defined as $f(h_m)$, the above formula can be written as:

$$P_{m,j}^{c} = \int_{H_{m,c}^{-}}^{H_{m,c}^{+}} f(h_m) dh_m \tag{2}$$

Therefore, the key to analyzing bus operation state is to determine the probability distribution of the bus headway. In reality, many random disturbance exist that will affect bus headways, such as traffic congestion, signals and bad weather. Therefore, different bus lines may have different headway distributions, and even stops on the same bus line may have different probability distributions. However, for the analysis of bus operation states along a given bus line, it is more meaningful to calibrate a relatively

uniform probability distribution. In other words, it is assumed that for the same line, most stations obey the same probability distribution. The following steps were developed to determine the probability distribution:

Step 1: list potential candidates of probability distribution function. According to the definition of the bus head time interval, its value range is $[0, +\infty)$. Therefore, the definition domain of the possible probability distribution must be the same as the value range of the bus headway. A list of common candidate distributions is listed in table 1 below:

[Insert Table 1 Candidate probability distributions Here]

Step 2. Estimate parameters involved in a possible probability distribution by the statistical parameter estimation methods with the headway samples. For all candidate probability distributions, the maximum likelihood estimation method (MLE) is used to estimate the unknown parameters.

Step 3. Test the Goodness-of-fit of different candidate probability distributions for each headway sample at each site of each line.

Step 4. From the results of the goodness-of-fit test, find the most suitable bus headway probability distribution. The significance level of 0.05 is adopted.

Dynamic evolution of bus operation states

As discussed in the state definition and classification section, dynamic evolution between different operation states can be observed, which means one state may transition to another state with certain probabilities. In reality, it can also be observed that without any intervention, the phenomenon of self-recovery will occur. This means that the generation and dissipation of bunching can happen during the operation process. In order to further understand the dynamic evolution between different states, this section uses Markov theory to analyze and predict the bus operation state.

In a time-series random process, one state can either change into another one or maintain unchanged. Each variation between two states is called transition, and the correlation probability of the variation is called transition probability. Such random process is called a Markov process if its immediate future state is only related to the current state and is independent from any other past states. In the Markov process, the mathematical probability method is used to calculate the transition probability between states in history.

Combined with the current state, the probability of occurrence of each state in the future can be accurately predicted. Its statistical expression is as follows (Chen et al. 2015):

Assuming sequence {X(t),t∈T} is a random process, T is the space of the time, and E is the space of the state. For any positive integer n ∈ E and any $t_1<t_2<\cdots<t_n$, t ∈ T, if $P\{X(t_1)=x_1, X(t_2)=x_2 \ldots\ldots, X(t_n)=x_n)>0$, the next equation must be satisfied:

$$P\{X(t_n)=x_n|X(t_1)=x_1 \ldots X(t_{n-1})=x_{n-1}\}=P\{X(t_n)=x_n|X(t_{n-1})=x_{n-1}\} \quad (3)$$

For the bus system, the stations can be regarded as a space of time, and the bus moving from station $n-1$ to n indicate a state transition. In this manuscript, it was assumed that the next station state was related only to the last station state which is reasonable as the previous states can be represented by the immediate past state and thus become irrelevant. As such, the state and time spaces of the bus operation Markov chain can be described as follow:

State space: C={1,2,3,4,5}, i.e. the five bus operation states.

Time space: Total number of stations for each bus line.

The one-step transition probability is the basis of the Markov chain analysis, and can be obtained by statistical analysis. When the sample size is sufficiently large, the one-step transition probability can be approximated by the frequency of state transfer, i.e. once the state threshold was calibrated, the bus operation state can be determined and then the transition probability of one state transitioning to another state can be calculated. It should be noted that state transition probability was also defined as route-specific. In this study, the one-step transition probability of bus route m can be calculated by equation 4 below.

$$p_{kl}^{m}=P\{X_{n+1}=l|X_n=k\}=\frac{F_{kl}^{m}}{F_{k}^{m}} \quad (4)$$

Where m_k represents the frequency of state k of route m in different time periods, m_{kl} is the frequency of state k transitioned to state l after one step transition, and p_{kl} is the one-step probability. If we represent the one-step transition probability in the form of a matrix, one-step transition probability matrix of route m ($P_{(1)}^{m}$) was obtained, and the n-step transition probability matrix of route m ($P_{(n)}^{m}$) can be calculated as follows:

$$P_{(n)}^{m} = [P_{(1)}^{m}]^{n} \tag{5}$$

Prediction of bus operation state

Multinomial logistic model is a statistical probability model that generalizes logistic regression problems to multiple class problems that may have two or more discrete outcomes (Hosomer et al. 1980). Given a set of independent variables, the state probability can be determined. The current headway and state in the next period were considered to be the independent and dependent variables, respectively. The logistic model is as given in Equation 6.

$$P\left(y = c_{m,j}^{i} \middle| h_{m,j-1}^{i}\right) = \frac{e^{\varepsilon_{m,c} + \beta_{m,c} * h_{m,j-1}^{i}}}{1 + \sum_{c=1}^{c=4} e^{\varepsilon_{m,c} + \beta_{m,c} * h_{m,j-1}^{i}}} \tag{6}$$

Where $h_{m,j-1}^{i}$ is the headway for the route m of ith trip at the stop $j-1$, and $c_{m,j}^{i}$ is the bus state for the bus route m of ith trip from stop $j-1$ to j. y is the dependent variable representing the bus operation state in next time period, which has five different outcomes. $\varepsilon_{m,c}$ and $\beta_{m,c}$ are the unknown parameters to be estimated.

Taking current bus headway as independent variable and the next-time bus running state as dependent variable, a multinomial logistic prediction model was developed. Once the current bus headway distance was obtained, the above formulas can be used to compute the probability of occurrence of each running state at the next moment. Based on the principle of maximum probability, the bus running state at the next moment can be determined.

Case Study

Case study setup

The data used in this study was obtained from the public transport company of Xi'an China, which covered over 280 bus routes. As of the end of 2017, the dataset included three data types: Global Positioning System (GPS) data, Automatic Vehicle Location (AVL) data and smart card data. The GPS data included route ID, longitude, latitude, arrival time and ID of vehicles. Among many bus routes, four

bus lines were selected for processing, namely #600, #601, #308 and #206. These four bus lines covered most urban areas of Xi'an and were located within a major bus corridor. The average daily numbers of GPS data points per line was about 200,000 to 400,000. Bus routes and important stations were highlighted in different colors in Figure 2 below.

[Insert Figure 2 Bus routes and main stations Here]

The headways were used to classify bus operation states and analyze the evolution among them. To provide an illustration of the compiled data, a descriptive summary was provided in Table 2.

[Insert Table 2 Descriptive of bus headways Here]

The bus operation states indicated the changes of headway between two consecutive buses. The GPS trajectories of eight buses from 7:00am to10:00am were studied and presented in Figure 3. The X axis was the station number, and the Y axis was the time of arrival at the station. The time difference between two consecutive buses at a same station was the bus headway. A higher distance between two adjacent trajectories indicated a larger headway between two consecutive buses, which meant the large interval state was more likely to be observed. On the contrary, when the headway between the two buses was relatively small, bus bunching state may occur. Generation and dissipation and of bus bunching can be observed from Figure 3. The overlap of two paths meant the two buses bunched together, which was an intuitive representation of bus bunching phenomenon. The phenomenon of bunching dissipation were shown in the figures that the two trajectories deviated from each other after overlapping. As seen from Figure 3, bus bunching was very common in the process of operation, and bus bunching generation and dissipation were alternating on the line. Two consecutive buses can start with a predetermined interval at the beginning, but in the course of operation random disturbances made their headway fluctuate. As a result the two buses would bunch together and then continue to run for a period of time in the state of bunching. Later on, the distance between them became wider and finally led to bunching dissipation. More details on bus bunching generation-dissipation process were presented in later sections.

[Insert Figure 3 Bus operation trajectories here]

Model Calibration

State classification

It should be stated that the thresholds varied among different lines, which was understandable as it was affected by many factors such as traffic conditions, length of bus route, number of stations, passenger demand and so on. In most existing literature, the threshold of bus bunching was given by experience. In this study, based on the bus GPS data, headway data was clustered and analyzed, and the bunching threshold was determined. Five days GPS data from different buses were randomly selected for analysis. It was found that the clustering centers of each state did not fluctuate much. Using route #600 as an example, the clustering center of the bunching state is distributed between 1.74 and 2.07, the bunching transition state is distributed between 4.78 and 5.5, the normal operation is distributed between 7.62 and 8.28, the large interval transition state is distributed between 11.02 and 12.03, and the large interval state is distributed between 15.03 and 16.29. As such, we took the mean value of the five-day clustering center as the final clustering center, so as to obtain a relatively robust state division for each line in table 3. The results showed that this division could adapt to different data of the same line, and the adaptability of each state reached more than 90%. It should also be noted that the modeling and prediction process in this manuscript was route-specific, i.e. for each route the parameters were calibrated separately (i.e. symbol m indicates different bus route in parameters p_{kl}^{m}, F_{kl}^{m}, $P_{m,j}^{c}$, $H_{m,c}^{+}$, $f(h_m)$ and other parameters throughout the manuscript). As such, the assumption of state thresholds of classification with K-means algorithm being stable and robust should be satisfied.

[Insert Table 3 Clustering result and state threshold Here]

Probability function calibration

The most common used methods for good-of-fit test include Kolmogorov-Smirnov (K-S) test and Chi-Squared test. As Chi-squared is more appropriate for large samples size, whereas the average number of sample data per day in our study was about 90-130. Therefore, K-S method was adopted to perform the goodness-of-fit test.

The good-of-fit test of each candidate probability distribution was carried out and the curing fitting results were shown in Table 4. The value in column “R” meant how many times the corresponding

distribution (in column "D") were rejected. A total of 6 distributions shown in Table 1 were tested for each bus route. Take route 600 as an example, it had 35 groups of headway sample data, and thus the total number of tests reached 35*6=210 times. The significance level (SL) of 0.05 was used as follows: due to randomness or other reasons, wrong judgment might be made and the probability of such wrong judgment was set to be 0.05. As such, for route 600, if the number of times a candidate distribution was rejected for less than 35*0.05=1.75 times, it was considered to have passed the test. The fitting results in Table 4 showed that Gama distribution outperformed other candidate distributions for route 600, 601 and 206, while Half-normal distribution was found to be the most appropriate distribution for route 308.

[Insert Table 4 Fitting result Here]

Result analysis

Estimation of state probabilities

With the probability function selected, the occurrence probability of bus operation states at each site can be calculated. Taking the bunching and the normal operation as an example, the results were shown in the Figure 4. A general collar-shaped trend in terms of the bunching probability can be observed. At the beginning of all four routes, the bus bunching probability was low, but it gradually increased as the bus went through more bus stations. The observation was consistent with our common understanding that bus bunching increased with the distance from the departure terminal, since the variability may increase as the bus travels further. The phenomenon of such sharp increase is also called bunching effect, meaning a lagging bus had to collect more passengers and therefor tended to fall further behind. On the other hand, another noteworthy finding was that the bus headway deterioration did not continue all the way to the last stop, rather, it recovered at the end. In all four subfigures, the bus bunching probability started low but increased as time went and reached a peak value sometime in the middle. After that, the bus bunching probability all decreased, indicating the bus operation conditions were improved with certain self-recovery mechanism.

[Insert Figure 4 Probability of bunching & normal operation Here]

The transition probability from one state into another during the bus operation process was calculated. Figure 5 presented the transition probability with the initial state being normal operation, and Figure 6 presented the transition probability with the initial state being bus bunching. It could be found that the transition probability from the initial state into others gradually increased while the bus travelled more bus stations. For example, in Figure 5 the transition probabilities from initial normal operation state (c=3) to other four states (c=1,2,4,5) were all increasing, which also meant the probability of remaining at normal operation state dropped (as the summation of all transitional probabilities equal to 1). The same observations can be made in Figure 6 as well, in which the initial state is bus bunching (c=1), and the transition probabilities to all other 4 states increased as the bus went through more bus stations, while the probability of keeping bunching decreased.

It should be noted that in this manuscript the bus operation state is assumed to be a regular stationary stochastic process. It can be found in Figures 5 and 6 that for all four studied routes, the transition probably at subsequent bus stops first increased significantly and then all became stable. The transition probability curves for scenarios that start with other states, i.e. large interval, large interval transition, keeping constant were also generated and similar trends were observed. In all these scenarios, the transition probability curve all increased at the beginning and then became stable and stopped fluctuating. As such, the assumption of bus operation state being regular stationary stochastic process should be valid.

[Insert Figure 5 Initial state is normal operation Here]

[Insert Figure 6 Initial state is bus bunching Here]

The above analysis demonstrated that the bus bunching phenomenon was a dynamic evolution process. During the operation process, the bus was transitioning between different states. In other words, bus bunching was a process that alternates between generating and dissipating. On one hand, bus bunching was not an irreversible phenomenon, meaning as the operation distance increased, the probability of maintaining bunching state became lower due to the increases of the disturbance introduced by many random factors. As a result, bunching dissipation phenomenon will occur at a certain moment. On the other hand, when the bunching state transferred to the other states, as the running distance increased, the probability of other states transferred (back) to the bunching state will become higher, so the bus may bunch together again. In

this process it can be found that one factor affected dissipation and generation of bus bunching was the number of bus stops.

In reality, when the number of bus stations in total was limited, if the bunching state occurred at some stations that were close to the terminal station, the possibility of dissipation of bunching may be low and sometimes the bus would remain in bunching state until it arrived at the terminal station. But in general, the bus operation was a dynamic process in which generation and dissipation occurred alternately on the entire bus route. Regardless of the state, the probability of maintaining current state unchanged during operation would be lower as the bus traveled. This was because during the operation of the bus, the disturbances caused by random factors was increasing. The operating state was more likely to be the same when the bus was not long into the operation, but as the running distance increased, the impact from disturbances became stronger, and the probability of running in its initial state would be lower.

State prediction accuracy analysis

The model estimation results and the evaluation of goodness-of-fit were shown in Table 5. The result showed the present headway had a significant effect on the state in which the bus arrives at the subsequent station. The independent variable was not removed from the model on the basis of low statistical significance (SIG.<0.05).

[Insert Table 5 Model estimation result Here]

The state probability under different headway of all 4 routes was shown in Figure 7 a-d, in which X axis was the headway while Y axis was the probability of each operation state. As the headway became larger, a sharp decrease can be observed in the probability of bus bunching in the next period, while the probability of other states became higher. The greater the current headway was, the more likely the next moment would be a large interval state. For bunching transition, normal operation and large interval tended to be coincident. A sharp increase was observed within certain range of headway for each state, followed by a sharp decrease beyond this range.

[Insert Figure 7 State conditional probability Here]

The area enclosed by the intersection of each curve was the prediction interval of each state. The state threshold is defined as the point where the conditional probability of two adjacent states is equal. For example, in figure 7-a, when red curve and green curve intersect at approximately 3.5s headway, the condition probabilities of these two states became equal and we define this particular point as the state threshold. As the multiple curves generated multiple intersection points, multiple state thresholds were obtained and they were used to define different bus operation states.

The prediction accuracy was also analyzed. The ground truth state was determined by the state threshold as described in Table 3, and was used to compared with the modeling output to calculate the prediction accuracy for each station, each state, and the overall accuracy. If the predicted state matched with the ground truth state, the result was said to be correct, otherwise it was said to be incorrect. The accuracy was then calculated by the percentage of times the model made correct prediction, and the result was shown in Figure 8 a-b.

In Figure 8a, it can be found that the developed model was able to achieve good prediction accuracy for all 4 bus routes, at each bus station. The overall accuracy rate was over 80%, which indicated satisfactory modeling outcomes. The bus route with the best prediction results was #600 (denoted in the red color), in which the prediction accuracy for all stations was above 75% except one, and with many stations above 90%. The bus route with least desired performance was #601 (denoted in green color), but still, the prediction of bus states at all stations was demonstrated to be higher than 70%, with over half stations higher than 80%.

Figure 8b presented the prediction accuracy for different operation states. X axis was the five operation states defined earlier, while x=6 indicated the overall prediction accuracy. Y axis was the accuracy probability with the same definition as Figure 8a. It can be found that the prediction accuracy varied for different states and different bus routes. For most routes (#600, #308 and #206), the prediction was the most accurate for normal operation state (with c=1), although the number was the lowest for bus route #601. The prediction accuracy for large interval state was relatively worse. For example for bus route #308, the accuracy was merely 64%, while this number ranged between 76% and 82% for the other three routes.

However, on average the prediction accuracy for all four routes were between 80% and 86%, indicating satisfactory modeling outcomes.

[Insert Figure 8 Predictive accuracy Here]

Conclusion

This study proposed a set of models and procedures to predict the bus operation state from the perspective of bus bunching. Headway was used as an independent variable to divide bus operation state into five categories: bus bunching, bus bunching transition, normal operation, large interval transition and large interval. K-means method was employed to classify the bus operation state in order to obtain the threshold of bus operation states. Markov chain model was developed to describe the state evolution between different bus operation states, and the state transition probabilities were calculated quantitatively. Finally, a multinomial logistic model was developed to predict the bus operation state. A case study was designed to test the performance of the proposed model, based on the GPS trajectory data collected from four bus routes in Xi'an, China. The results showed that the proposed model was able to accurately predict the bus operation states.

The present work might be further strengthened in a few directions. In this manuscript, a bus route was considered to be operating independently, and thus the impact of overlap between multiple bus routes to the evolution of bus operation states was not considered. The modeling of multiple buses arriving and leaving a single bus station would require more microscopic analysis and simulation, and the modeling of multiple buses competing for the limited amount of bus stations resources.

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Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request (including the network, GPS trajectories data, and the modeling outputs of the four bus routes).

References

Andres, M., and R. Nair. 2017. "A predictive-control framework to address bus bunching." *Transportation Research Part B: Methodological*. 104: 123-148. https://doi.org/10.1016/j.trb.2017.06.013.

Borgwardt, S., A. Brieden, and P. Gritzmann. 2017. "An LP-based K-means algorithm for balancing weighted point sets." *European Journal of Operational Research*. 263 (2): 349-355. https://doi.org/10.1016/j.ejor.2017.04.054.

Chen, C., W. Liu, L. Chao-Hsin, and Q. Chen. 2015. "A markov chain model for predicting transient particle transport in enclosed environments." *Building & Environment*. 90: 30-36. https://doi.org/10.1016/j.buildenv.2015.03.024.

Chica-Olmo, J., and H. Gachs-Sánchez. 2018. "Route effect on the perception of public transport services quality." *Transport Policy*. 67: 40-48. https://doi.org/10.1016/j.tranpol.2017.03.024.

Cortés, C. E., J. Gibson, A. Gschwender A, and M. Munizaga. 2011. "Commercial bus speed diagnosis based on GPS-monitored data." *Transportation Research Part C:* Emerging *Technologies*. 19 (4): 695-707. https://doi.org/10.1016/j.trc.2010.12.008.

De Oña, R., G. López, F. J. D. de los Rios F J D, and J. de Oña. 2014. "Cluster analysis for diminishing heterogeneous opinions of service quality public transport passengers." *Procedia-Social and Behavioral Sciences*. 162: 459-466. https://doi.org/10.1016/j.sbspro.2014.12.227.

Deng, Y, X. Liu, X. Hu, M. Zhang. "Reduce bus bunching with a real-time speed control algorithm considering heterogeneous roadway conditions and intersection delays." ASCE Journal of Transportation Engineering, Part A: Systems, in press.

Ding, Y., and S. I. Chien. 2001. "Improving transit service quality and headway regularity with real-time control." *Transportation Research Record.* 1760 (1): 161-170. https://doi.org/10.3141/1760-21.

Dimitrov, S., A. Ceder A, S. Chowdhury, and M. Monot. 2017. "Modeling the interaction between buses, passengers and cars on a bus route using a multi-agent system." *Transportation Planning and Technology*. 40 (5): 592-610. https://doi.org/10.1080/03081060.2017.1314504.

El‐Geneidy, A. M., J. Horning, and K. Krizek. 2011. "Analyzing transit service reliability using detailed data from automatic vehicular locator systems." *Journal of Advanced Transportation*. 45 (1): 66-79. https://doi.org/10.1002/atr.134.

Feng, W., and F. Migule. 2011. "Empirical analysis of bus bunching distributions and attributes using archived AVL/APC bus data." *11th International Conference of Chinese Transportation Professionals (ICCTP)*. 4330-4341. http://dx.doi.org/10.1061/41186(421)427.

Furth, P. G., and T. Muller. 2007. "Service reliability and optimal running time schedules." *Transportation Research Record*. 1 (2034): 55-61. https://doi.org/10.3141/2034-07.

Hammerle, M., M. Haynes, and S. McNeil. 2005. "Use of automatic vehicle location and passenger count data to evaluate bus operations." *Transportation Research Record.* 1 (1903): 27-34. https://doi.org/10.1177/0361198105190300104.

Hans, E., N. Chiabaut N, L. Leclercq, and R. Bertini. 2015. "Real-time bus route state forecasting using particle filter: an empirical data application." *Transportation Research Procedia*. 6: 434-447. https://doi.org/10.1016/j.trpro.2015.03.033.

Hosmer, D. W., and S. Lemesbow. 1980. "Goodness of fit tests for the multiple logistic regression model." *Communications in statistics-Theory and* Methods. 9 (10): 1043-1069. https://doi.org/10.1080/03610928008827941 .

Hu, J. H., and J. Liang. 2017. "Inferring the travel states of bus rapid transit between neighboring stations within a bayesian framework." *Transportation Systems Engineering and Information Technology.* 17 (2): 126-135. http://dx.doi.org/10.16097/j.cnki.1009-6744.2017.02.019.

Li, J., X. Chen, X. Li, and X. Guo. 2013. "Evaluation of public transportation operation based on data envelopment analysis." *Procedia-Social and Behavioral Sciences*. 96:148-155. https://doi.org/10.1016/j.sbspro.2013.08.020 .

Mazloumi, E., G. Currie, and M. Sarvi. 2008. "Assessing measures of transit travel time variability and reliability using AVL data.". 2008. Washington, DC: Transportation Research Board.

Moreira-Matias, L., J. Gama, J. Mendes-Moreira J, and J. Freire de Sousa. 2014. "An incremental probabilistic model to predict bus bunching in real-time." *Advances In Intelligent Data Analysis Xiii*. 8819: 227-238. http://dx.doi.org/10.1007/978-3-319-12571-8_20.

Ruan, M. 2009. "Probability-based bus headway regularity measure." *Intelligent Transport Systems*. 3 (4): 400-408. https://doi.org/10.1049/iet-its.2008.0088 .

Ruan, M., and J. Lin. 2009. "An investigation of bus headway regularity and service performance in Chicago bus transit system." .2009. Chicago, USA: Transport Chicago 2009 Annual Conference.

Sirmatel, I. I., and N. Geroliminis. 2017. "Dynamical modeling and predictive control of bus transport systems: a hybrid systems approach." *IFAC-PapersOnLine*. 50 (1): 7499-7504. https://doi.org/10.1016/j.ifacol.2017.08.1067.

Sirmatel, I. I., and N. Geroliminis.2018. "Mixed logical dynamical modeling and hybrid model predictive control of public transport operations." *Transportation Research Part B: Methodological*. 114: 325-345. https://doi.org/10.1016/j.trb.2018.06.009.

Strathman, J. G., J. Kimpel, J. Dueker, L. Richard and C. Steve. 2002. "Evaluation of transit operations: data applications of Tri-Met's automated bus dispatching system." *Transportation*. 29 (3): 21-345. http://dx.doi.org/10.1023/A:1015633408953.

Song, X., T. Jin, L. D. Li, and S. Huang. 2013. "Travel time-based approach for division and recognition of single-line bus transit running state." .2013. *4th International Conference on Transportation Engineering*. https://doi.org/10.1061/9780784413159.290

Transportation Research Board. 2013. "Transit capacity and quality of service manual, Third Edition".

Varga, B., T. Tettamanti, and B. Kulcsár. 2018. "Optimally combined headway and timetable reliable public transport system." *Transportation Research: Part C: Emerging Technologies*. 92:1-26. https://doi.org/10.1016/j.trc.2018.04.016 .

Wang, K., D. Guo, and L. U. Guoshen.2017. "Evaluation and analysis of general public transportation operation: a case study in Urumqi." *Urban Public Transport*. (01): 19-24.

Wang, T. "A flexifare bus (F2B) system to reduce bus bunching problem." *International Conference on Progress in Informatics and Computing*. 2014: 651-655. https://doi.org/10.1109/pic.2014.6972415 .

Watkins, K. E, B. Ferris, A. Borning, G. Rutherford and D. Layton. 2011. "Where is my bus? impact of mobile real-time information on the perceived and actual wait time of transit riders." *Transportation Research Part A: Policy and Practice*. 45 (8): 839-848. https://doi.org/10.1016/j.tra.2011.06.010 .

Xiao, L., and L. Cheng. "State classification algorithm for bus based on hierarchical support vector machine." *.2015. 8th International Symposium on Computational Intelligence and Design (ISCID)*. 2: 649-652. https://doi.org/10.1109/iscid.2015.259 .

Yan, Y., Z. Liu, and Y. Bie. 2016. "Performance evaluation of bus routes using automatic vehicle location data." *Journal of Transportation Engineering*. 142 (8): 04016029. https://doi.org/10.1061/(asce)te.1943-5436.0000857 .

Yuan, T., Shi. T, and M. LI. 2015. "The bus operating condition discriminant algorithm based on GPS data." *Road Traffic & Safety.* 2:005. http://dx.doi.org/10.13986/j.cnki.jote.2015.02.005.

Yu, H., D. Chen, Z. Wu, X. Ma and Y. Wang. 2016. "Headway-based bus bunching prediction using transit smart card data." *Transportation Research Part C: Emerging Technologies*. 72: 45-59. https://doi.org/10.1016/j.trc.2016.09.007.

Zhang, J., J. Ding, J. Long, and Q. Shi. 2015. "The exploration of time-headway characteristic and operation status on the bus route." *Transportation Systems Engineering and Information Technology*. 15 (6): 220-226. http://dx.doi.org/10.16097/j.cnki.1009-6744.2015.06.033.

Zhang, S., X. Yang, and J. Teng. 2011. "Evaluation of bus service reliability based on AVL information." *ICCTP 2011: Towards Sustainable Transportation Systems*. 2890-2899. https://doi.org/10.1061/41186(421)287 .

Table 1 Candidate probability distributions

Distribution	Probability density function	Unknown parameters	Estimation method
Half-normal	$f(x\|\sigma) = \frac{\sqrt{2}}{\sigma\sqrt{\pi}} e^{-\frac{x^2}{2\sigma^2}}$	σ	MLE
Log-logistic	$f(x\|\alpha,\beta) = \frac{\alpha x^{\alpha-1}}{\beta^\alpha}[1+\left(\frac{x^\alpha}{\beta^\alpha}\right)]^{-2}$	α , β	MLE
Gamma	$f(x\|\alpha,\beta) = \frac{x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} e^{-\frac{x}{\beta}}$	α , β	MLE
Weibull	$f(x\|\alpha,\beta) = \frac{\alpha}{\beta}(\frac{x}{\beta})^{\alpha-1}\exp\left(-\left(\frac{x}{\beta}\right)^\alpha\right)$	α , β	MLE
Log-normal	$f(x\|\mu,\sigma) = \frac{1}{x\sigma\sqrt{2\pi}}\exp\left(-\left(\frac{lnx-\mu}{\sqrt{2\sigma}}\right)^2\right)$	μ,σ	MLE
Erlang	$f(x\|\beta,m) = \frac{x^{m-1}}{\beta^m \Gamma(m)} e^{-\frac{x}{\beta}}$	β,m	MLE

Table 2 Descriptive of bus headways

Route ID	#600	#601	#308	#206
Length (km)	25	18.9	16.8	13
Number of Stops	35	30	30	21
Average Scheduled headway (min)	7.5	6	5	6
Number of headways Per Day	2,905	3,540	3,360	1,953
Number of headways Per Stop	83	118	112	93

Table 3 Clustering result and state threshold

Route	1	2	3	4	5
600	(0,3.484]	(3.484,6.73]	(6.73,9.81]	(9.81, 13.51]	(13.51, +∞)
601	(0,3.160]	(3.160,6.11]	(6.11,9.43]	(9.43, 13.94]	(13.94, +∞)
308	(0,3.499]	(3.499,6.99]	(6.99,11.2]	(11.2, 15.61]	(15.61, +∞)
206	(0,3.854]	(3.854,7.48]	(7.48,11.3]	(11.3, 15.37]	(15.37, +∞)

Table 4 Fitting result

#600		#601		#308		#206	
D	R	D	R	D	R	D	R
Half-normal	24	Half-normal	10	Half-normal	1	Half-normal	8
Log-logistic	4	Log-logistic	30	Log-logistic	25	Log-logistic	20
Gamma	0	Gamma	1	Gamma	11	Gamma	0
Weibull	12	Weibull	30	Weibull	10	Weibull	19
Log-normal	33	Log-normal	28	Log-normal	19	Log-normal	20
Erlang	35	Erlang	29	Erlang	25	Erlang	17

Table 5 Model estimation result

Parameter		ε	Wald	β	Wald
Route	State				
600	1	76.881***	724.141	-9.714***	846.026
	2	66.051***	589.654	-6.849***	727.939
	3	48.783***	727.939	-4.324***	360.287
	4	24.146***	123.551	-1.793***	115.837
601	1	19.919***	807.291	-2.39***	1131.076
	2	16.946***	612.842	-1.6***	668.792
	3	11.493***	313.106	-0.887***	271.925
	4	6.418***	119.042	-0.442***	96.841
308	1	21.346***	623.445	-2.306***	1036.759
	2	18.392***	427.483	-1.552***	595.229
	3	14.272***	301.45	-1.002***	310.006
	4	9.131***	142.958	-0.559***	127.05
206	1	41.622***	474.303	-4.859***	696.292
	2	35.713***	365.832	-3.248***	462.569
	3	27.272***	233.457	-2.083***	254.072
	4	16.759***	110.894	-1.118***	107.85

1) *** significant at 99%

2) For both routes, large interval state is regarded as the reference category.